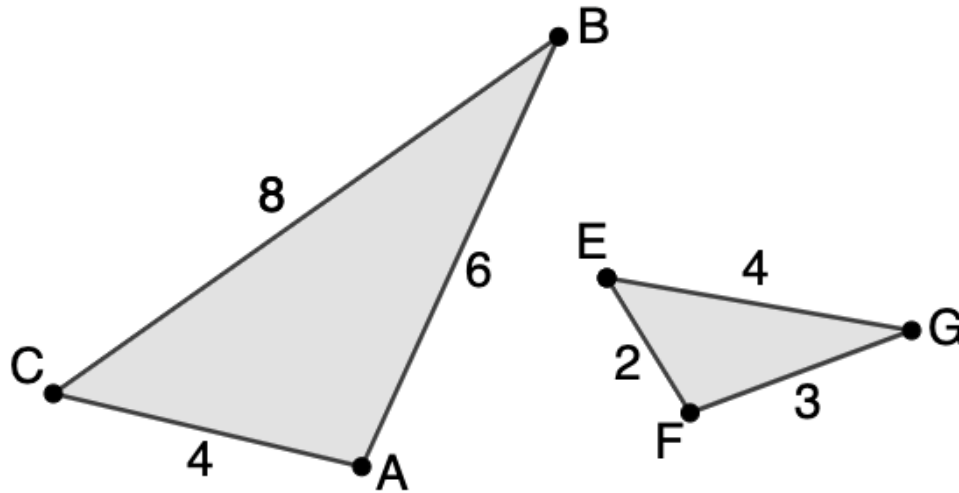


Triangle ABC and FGE are shown.



1. Which segment corresponds with segment AB ?

Segment FG

2. Which segment corresponds with segment BC ?

Segment GE

3. Which segment corresponds with segment AC ?

Segment FE

4. Is there a constant of proportionality that exists with these triangles? If so, what is it?

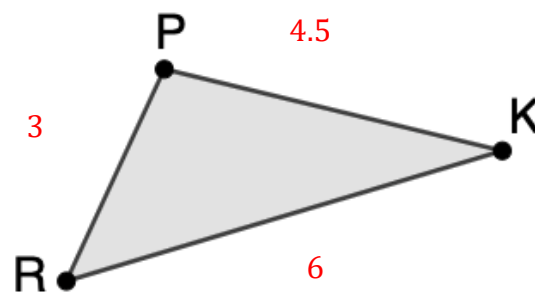
Yes, $\frac{1}{2}$ or 2

5. Are these two triangles similar? Why or why not?

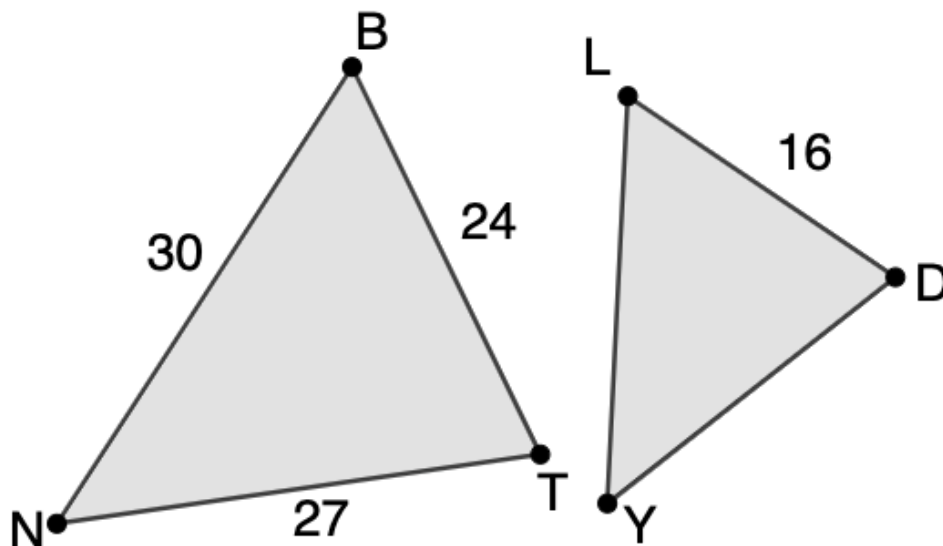
Yes, because the corresponding sides are proportional.

6. Label triangle PKR so that it is similar to triangle FGE .

Answers will vary



Triangle NBT is similar to triangle YLD .



7. Which segment corresponds with segment LD ?

Segment BT

8. What is the constant of proportionality between triangle NBT and triangle YLD ?

$$\frac{16}{24} = \frac{2}{3} \quad \text{or} \quad \frac{24}{16} = \frac{3}{2}$$

9. What is the length of side YD ?

$$27 \cdot \frac{2}{3} = \boxed{18}$$

10. What is the length of side LY ?

$$30 \cdot \frac{2}{3} = \boxed{20}$$